

HDOC 19

Question:

The algorithm for calculating the Semester Grade Point Average (SGPA) and Cumulative Grade Point Average

(CGPA) for 'm' semesters is given below. Each semester has 'n' courses, where 'n' may vary between semesters.

Assume that the student has passed all courses. Prepare a test-case table covering valid, invalid, and boundary

inputs; write the C program based on the given algorithm; compile and execute it; and test it against the prepared

test cases.

Task1: Algorithm:

1. Start.
2. Define the variables and their types: global_id, m, n, semester, course, code, credit, GP, semester_TC, semester_TGC, cumulative_TC, and cumulative_TGC as integers; grade as a character; and SGPA and CGPA as floating-point variables.
3. Get global_id.
4. Validate global_id: If global_id is not positive, display an appropriate error message and request it again. Repeat until a positive value is entered.
5. Get m.
6. Validate m: If m is not positive, display an appropriate error message and request it again. Repeat until a positive value is entered.
7. Set the cumulative values: cumulative_TC = 0, cumulative_TGC = 0
8. Repeat Steps 8.1 to 8.8 for m semesters:
 - 8.1. Set the semester totals:
semester_TC = 0, semester_TGC = 0
 - 8.2. Get n.

8.3. Validate n: If n is not positive, display an appropriate error message and request it again. Repeat until a positive value is entered.

8.4. Repeat Steps 8.4.1 to 8.4.9 for n courses:

8.4.1. Get code.

8.4.2. Validate code: If it is not positive, display an error message and request it again. Repeat until a positive value is entered.

8.4.3. Get credit.

8.4.4. Validate credit: If it is not positive, display an error message and request it again. Repeat until a positive value is entered.

8.4.5. Get grade.

8.4.6. Validate grade: Accept only O, A, B, C, D, E, or P. If the grade is invalid, display an error message and request it again. Repeat until a valid grade is entered.

8.4.7. Find GP corresponding to grade using the university's conversion scale:

O = 10, A = 9, B = 8, C = 7, D = 6, E = 5, P = 4

8.4.8. Calculate the semester total grade credits:

$\text{semester_TGC} = \text{semester_TGC} + (\text{GP} * \text{credit})$

8.4.9. Calculate the semester total credits:

$\text{semester_TC} = \text{semester_TC} + \text{credit}$

8.5. Calculate the SGPA of the current semester:

$\text{SGPA} = \text{semester_TGC} / \text{semester_TC}$

8.6. Display semester and SGPA.

8.7. Add the semester grade credits to the cumulative grade credits:

$\text{cumulative_TGC} = \text{cumulative_TGC} + \text{semester_TGC}$

8.8. Add the semester credits to the cumulative credits:

$\text{cumulative_TC} = \text{cumulative_TC} + \text{semester_TC}$

9. Calculate the CGPA:

$\text{CGPA} = \text{cumulative_TGC} / \text{cumulative_TC}$

10. Display global_id, m, cumulative_TC, and CGPA.

11. Stop

Task2: Test case table:

Test Case	Scenario	Input	Expected Output	Actual Output	Result
TC-01	Valid: 1 semester, 3 courses	ID=101, m=1; Courses: (1011,4,O), (1012,3,A), (1013,2,B)	SGPA = 9.22, CGPA = 9.22, Total Credits = 9	SGPA = 9.22, CGPA = 9.22, Total Credits = 9	Pass
TC-02	Valid: 2 semesters with different courses	ID=102, m=2; Sem-1: (201,4,A), (202,3,B); Sem-2: (301,4,O), (302,2,C), (303,3,A)	Sem-1 SGPA = 8.57, Sem- 2 SGPA = 9.00, CGPA = 8.81, Credits = 16	Same as expected	Pass
TC-03	Boundary: minimum positive values	ID=1, m=1, n=1; Course (1,1,P)	SGPA = 4.00, CGPA = 4.00, Credits = 1	Same as expected	Pass
TC-04	Boundary: highest grade O	ID=103, m=1, n=1; (401,1,0)	SGPA = 10.00, CGPA = 10.00	Same as expected	Pass
TC-05	Invalid Global ID followed by valid ID	ID=-10, then 104	Error for -10; accepts 104	Error displayed and 104 accepted	Pass
TC-06	Invalid number of semesters	ID=105, m=0, then 1	Error for m=0; accepts 1	Error displayed and 1 accepted	Pass
TC-07	Invalid number of courses	n=-2, then 1	Error for -2; accepts 1	Error displayed and 1 accepted	Pass
TC-08	Invalid course code	code=0, then 501	Error for 0; accepts 501	Error displayed and 501 accepted	Pass
TC-09	Invalid credit	credit=0, then 4	Error for 0; accepts 4	Error displayed and 4 accepted	Pass

Task3: C Program:

```
#include <stdio.h>
int main()
{
    int global_id, m, n;
    int semester, course;
    int code, credit, GP;
    int semester_TC, semester_TGC;
    int cumulative_TC, cumulative_TGC;
    char grade;
    float SGPA, CGPA;

    /* Input and validation of Global ID */
    do
    {
        printf("Enter Global ID: ");
        scanf("%d", &global_id);

        if (global_id <= 0)
            printf("Error: Global ID must be positive.\n");
    }
    while (global_id <= 0);

    /* Input and validation of number of semesters */
    do
    {
        printf("Enter number of semesters: ");
        scanf("%d", &m);
        if (m <= 0)
            printf("Error: Number of semesters must be positive.\n");
    }
    while (m <= 0);
    cumulative_TC = 0;
    cumulative_TGC = 0;

    /* Semester loop */
    for (semester = 1; semester <= m; semester++)
    {
```

```
semester_TC = 0;
semester_TGC = 0;
printf("\n--- Semester %d ---\n", semester);
```

```
/* Input and validation of number of courses */
```

```
do
{
    printf("Enter number of courses: ");
    scanf("%d", &n);
    if (n <= 0)
        printf("Error: Number of courses must be positive.\n");
}
while (n <= 0);
```

```
/* Course loop */
```

```
for (course = 1; course <= n; course++)
{
    printf("\nCourse %d\n", course);
```

```
/* Course code */
```

```
do
{
    printf("Enter course code: ");
    scanf("%d", &code);
    if (code <= 0)
        printf("Error: Course code must be positive.\n");
}
while (code <= 0);
```

```
/* Course credit */
```

```
do
{
    printf("Enter credit: ");
    scanf("%d", &credit);
    if (credit <= 0)
        printf("Error: Credit must be positive.\n");
}
while (credit <= 0);
```

```
/* Grade */
```

```
do
```

```

{
    printf("Enter grade (O/A/B/C/D/E/P): ");
    scanf(" %c", &grade);
    switch (grade)
    {
        case 'O':
            GP = 10;
            break;
        case 'A':
            GP = 9;
            break;
        case 'B':
            GP = 8;
            break;
        case 'C':
            GP = 7;
            break;
        case 'D':
            GP = 6;
            break;

        case 'E':
            GP = 5;
            break;
        case 'P':
            GP = 4;
            break;
        default:
            GP = 0;
            printf("Error: Invalid grade. Enter again.\n");
    }

}

while (GP == 0);
semester_TGC = semester_TGC + (GP * credit);
semester_TC = semester_TC + credit;
}
SGPA = (float)semester_TGC / semester_TC;
printf("\nSemester %d SGPA = %.2f\n", semester, SGPA);
cumulative_TGC = cumulative_TGC + semester_TGC;
cumulative_TC = cumulative_TC + semester_TC;

```

```

}
CGPA = (float)cumulative_TGC / cumulative_TC;
printf("\n===== FINAL RESULT =====\n");
printf("Global ID      : %d\n", global_id);
printf("Total Semesters  : %d\n", m);
printf("Cumulative Credits : %d\n", cumulative_TC);
printf("CGPA           : %.2f\n", CGPA);
return 0;
}

```

Task4: Compilation and Execution:

```

(kali㉿kali)-[~/sgpa]
└─$ ls
sgpa.c
(kali㉿kali)-[~/sgpa]
└─$ gcc sgpa.c -o sgpa
(kali㉿kali)-[~/sgpa]
└─$ ./sgpa
Enter Global ID: 101
Enter number of semesters: 1
--- Semester 1 ---
Enter number of courses: 3
Course 1
Enter course code: 1011
Enter credit: 4
Enter grade (O/A/B/C/D/E/P): 0
Course 2
Enter course code: 1012
Enter credit: 3
Enter grade (O/A/B/C/D/E/P): A
Course 3
Enter course code: 1013
Enter credit: 2
Enter grade (O/A/B/C/D/E/P): B
Semester 1 SGPA = 9.22

===== FINAL RESULT =====
Global ID      : 101
Total Semesters : 1
Cumulative Credits : 9
CGPA           : 9.22

```